

Lab program-15:

15. Design a C program to organise the file using a two level directory structure.

Code:

```
#include <stdio.h>

#define MAX_USERS 10
#define MAX_FILES 10

int main()
{
    char users[MAX_USERS][30];
    char files[MAX_USERS][MAX_FILES][50];

    int numberOfUsers;
    int numberOfFiles[MAX_USERS];

    printf("Enter number of users: ");
    scanf("%d", &numberOfUsers);

    for (int i = 0; i < numberOfUsers; i++)
    {
        printf("\nEnter name of User %d: ", i + 1);
        scanf("%s", users[i]);

        printf("Enter number of files for %s: ", users[i]);
        scanf("%d", &numberOfFiles[i]);

        printf("Enter file names:\n");

        for (int j = 0; j < numberOfFiles[i]; j++)
        {
            scanf("%s", files[i][j]);
        }
    }

    printf("\n\nTwo Level Directory Structure\n");

    printf("\nMaster File Directory:\n");

    for (int i = 0; i < numberOfUsers; i++)
    {
        printf("\nUser Directory: %s\n", users[i]);

        for (int j = 0; j < numberOfFiles[i]; j++)
```

```

        {
            printf("    %s\n", files[i][j]);
        }
    }

    return 0;
}

```

OUTPUT:

```

● PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_15.exe
Enter number of users: 3

Enter name of User 1: U1
Enter number of files for U1: 3
Enter file names:
songs.txt
unreleased.txt
upcoming.txt

Enter name of User 2: harris
Enter number of files for harris: 2
Enter file names:
bangers.txt
unreleased.txt

Enter name of User 3: ani
Enter number of files for ani: 1
Enter file names:
joni.txt

Two Level Directory Structure

Master File Directory:

User Directory: U1
    songs.txt
    unreleased.txt
    upcoming.txt

```